

Week 15 Assignment — Capstone Presentation Preparation

CE 310 | Groups of 3 | Presentation in Week 16

Your Week 16 presentation is worth 60 points of your capstone grade. This document describes what to prepare. The grading rubric is in `Week15_Presentation_Rubric.md`.

How to Submit

☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	PDF — export your slides to PDF; combine with the one-page prep sheet into a single PDF (or submit as two files)
Team submission	One team member submits on behalf of the whole team. List all three names on the title slide.
File name	lastname1_lastname2_lastname3_week15_slides.pdf (alphabetical by last name)
Submission	Upload to D2L Dropbox at least 24 hours before your assigned presentation session (Mon Dec 7 or Wed Dec 9)

What You Are Delivering

A 4-slide, 4-minute presentation (+ 1 minute Q&A) to your classmates and the instructor, analyzing your team's chosen dataset. Imagine you are a junior engineering consultant presenting findings to your dataset's client — be concise; with every team presenting across two class days, there's no time to spare.

You completed the analysis in the capstone lab notebook. This assignment is the presentation preparation — your slides, speaking notes, and a brief one-page prep sheet (submitted on D2L before class).

Slide Requirements

Slide	Title	Must include
1	Dataset & Client Context	Dataset name/source, row count (ANSWER_A1), mean X and Y (ANSWER_A2/A3), a profile figure
2	Regression Findings	Model 1 vs Model 2 comparison table; X slope (ANSWER_B1) with units; GROUP coefficient (ANSWER_B2) with physical interpretation
3	Group Comparison Analysis	T-test result; Group A mean vs Group B mean (ANSWER_C1 and the other group's mean); explain why t-test and regression may give different pictures
4	Uncertainty & Recommendation	Baseline estimate (ANSWER_D1); MC P10–P90 range; one specific, quantified recommendation

A 5th slide is optional for any additional analysis (subgroup deep-dive, alternative variable definition, residual diagnostics, etc.).

Submission on D2L (due 24 hours before your session)

Submit a PDF of your slides plus a one-page “prep sheet” with: 1. Your dataset, client, and the numbers from each ANSWER_ variable 2. Two sentences on your most interesting finding 3. One anticipated Q&A question and your planned answer

The prep sheet is graded on completion (2 pts extra credit), not quality.

Group Work Expectations

All three group members must speak during the presentation. Divide the slides so each person presents at least one slide. During Q&A, any group member may answer.

You will be asked at least one question comparing your findings to another team's topic (e.g., "Your GROUP coefficient was negative — did you find anything similar to other teams' counterintuitive results?"). Prepare to discuss your methodology in terms other teams — working in a completely different domain — can follow.

Presentation Tips

- Quantify everything. "The GROUP coefficient is 3.5 units per hour" is better than "the two groups differ."
- Connect to the client. Your slides should explain what the numbers mean for your specific stakeholder, not just report them.
- If your GROUP coefficient's sign is counterintuitive, that's usually your most compelling finding — explain the mechanism, don't hide it.
- If your coefficient seems unusually large, check for confounding (does your grouping variable overlap with something seasonal or cyclical in your predictor?) and explain what you found either way.
- Small coefficients are not bad results. A weak relationship can still be a real, useful finding — explain what it means for your client's decision.